

# Sathwik Mohan Vallala

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## SUMMARY

Cybersecurity professional with 5+ years of experience in breaking and patching business critical products and a Certified Ethical Hacker (CEH) certification, specializing in offensive and defensive security. Experienced in penetration testing, attack surface analysis, vulnerability management, and security automation across multiple products. Proficient with industry-standard tools and frameworks, including Maltego and CVSS, and skilled at developing automation to streamline vulnerability triage and investigation. Passionate about leveraging AI and automation to improve security workflows, reduce manual effort, and enable faster risk mitigation while collaborating closely with engineering teams to deliver secure products. Committed to staying current with emerging threats, evolving attack techniques, and the latest advancements in cybersecurity to proactively strengthen security posture.

## EXPERIENCE

### Product Security Engineer

Jun. 2024 – Jul. 2026

*Omnissa, LLC*

*Herndon, VA*

- Integrated AI into application security workflows to reduce manual effort, accelerating remediation of 50+ critical vulnerabilities prioritized using CVSS.
- Combined AI-assisted analysis with manual security expertise to evaluate attack surfaces for 30+ customer features, delivering secure implementation recommendations.
- Enhanced penetration testing efficiency using AI for reconnaissance, payload generation, and vulnerability triage across 10+ product releases.
- Automated portions of secure code review using AI, increasing review speed while maintaining high-confidence findings across 20+ releases.
- Used AI to accelerate threat modeling, rapidly identifying attack paths and validating mitigations for 10+ major engineering initiatives.
- Scaled secure code review efforts with AI-assisted static analysis and code understanding, proactively identifying risks across 75% of product codebases.
- Communicated complex security issues, improving security awareness and alignment across 5+ cross-functional teams.
- Initiated a monthly cybersecurity round table to keep myself and the team up-to-date with events in cyber-security world.

### Product Security Engineer (CarbonBlack)

Nov. 2023 – Jun. 2024

*Broadcom Inc.*

*Herndon, VA*

- Reduced potential attack surface by identifying & resolving critical security issues measured by CVSS scoring systems.
- Executed penetration testing, uncovering & exploiting vulnerabilities to strengthen product security.
- Applied threat modeling techniques to identify potential attack vectors & guide secure design decisions within engineering teams.
- Proactively detected & mitigated potential security risks during secure code reviews, preventing exploitation.
- Communicated complex technical security issues fostering understanding & buy-in from product teams.

### Security Response Engineer

Apr. 2022 – Nov. 2023

*VMWare Inc.*

*Herndon, VA*

- Analyzed incoming security vulnerability reports, assessed urgency, & ensured timely responses.
- Validated & assessed reported vulnerabilities, assigned severity levels (CVSSv3.1), & documented exploitation steps & mitigation strategies.
- Coordinated vulnerability communication across internal teams & planned customer communication.
- Collaborated with researchers for responsible vulnerability disclosure & published VMware Security Advisories (VMSA) to inform customers about required actions.
- Defined & implemented corrective actions like mitigations, IOCs, workarounds, remediations to address vulnerabilities & minimize customer risk.
- Proactively monitored for exploit activity & updated advisories to reflect the evolving threat landscape.
- Streamlined security processes by developing & implementing automations, enhancing team efficiency & effectiveness.

### Digital Forensic Researcher & Developer

Dec. 2020 – Mar. 2022

*ADF Solutions Inc*

*Reston, VA*

- Ensured accuracy & reliability of existing digital artifact collection modules through rigorous testing & verification procedures.
- Independently researched & developed new digital artifact collection modules, expanding the capabilities of the system.
- Demonstrated creativity & problem-solving skills by devising innovative solutions for collecting diverse digital artifacts.
- Demonstrated strong team spirit & adaptability by actively assisting colleagues & fulfilling diverse project requirements.
- Leveraged Python scripting expertise to create custom plugins for efficient parsing & extraction of various digital artifacts & evidence, streamlining investigative processes.
- Maintained a high level of quality assurance by conducting thorough testing & validation of developed plugins before integration into the operational DEI Tool.

## Trainee Security Analyst - Internship

Jul. 2019 – Oct. 2019

*SOC Experts*

*Bangalore, India*

- Bolstered security with diverse SIEM tools (Splunk, QRadar, LogRhythm) for threat detection & response.
- Analyzed & monitored logs to uncover suspicious activity & potential incidents.
- Conducted basic digital forensics, collecting & analyzing evidence for investigations.
- Thrived in simulated SOC, collaborating effectively to identify & respond to security threats.

## Security Analyst - Internship

Jun. 2017 – Jun. 2018

*Berry9 IT Services Pvt Ltd*

*Hyderabad, India*

- Mastered ethical hacking through personal projects, fueled by passion.
- Identified vulnerabilities in real-world healthcare pen testing project.
- Earned CEH certification, validating ethical hacking expertise.

## EDUCATION

### George Mason University

*Master's of Science in Digital Forensics & Cyber Analysis - 3.89/4.0*

*Fairfax, VA*

### Jawaharlal Nehru Technological University

*Bachelor's of Technology in Computer Science Engineering - 4.0/4.0*

*Hyderabad, India*

## PROJECTS

### Collaborative ToDo | *Python, Forensics, Evidence*

Apr. 2020 – May 2020

- Vibe Coded with AI
- Self hosted privacy
- Password less login
- A ToDo where you can collaborate with others

### Autopsy WhatsApp Plugin | *Python, Forensics, Evidence*

Apr. 2020 – May 2020

- Enhanced *Autopsy* capabilities by independently developing a new plugin, demonstrating expertise in forensics tool customization & automation.
- Recovered & parsed digital evidence from the *WhatsApp* application on *Windows* systems, utilizing advanced data extraction & parsing techniques.

### Chat over network with Python | *Python, Network, Encryption*

Jun. 2018 – Jul. 2018

- Built a privacy-forward chat application with strong cryptography, empowering secure communication.
- Implemented robust end-to-end encryption for secure communication, ensuring message confidentiality & protecting user data.
- Enabled user control & independence by designing a self-hostable chat server, fostering privacy & data sovereignty.

### File Sharing | *PHP, HTML, SQL, Network*

Mar. 2018 – Apr. 2018

- Streamlined internal file sharing with a user-friendly platform designed for specific network needs.
- Prioritized security by implementing password authentication to restrict unauthorized access to shared files.

## CERTIFICATIONS & AWARDS

Security Code Review Ninja - 2nd Degree Black Belt | *Secure Coding Dojo*

December 2023

Digital Evidence Investigator | *ADF Solutions*

December 2020

Certified Autopsy User | *Basis Technology*

October 2020

Certified Ethical Hacker | *EC Council*

October 2017

## TECHNICAL SKILLS

### Professional

- LLM's
- Vulnerability Assessment
- Penetration Testing
- SAST
- Log Analysis
- Forensic Analysis
- Memory Forensics
- SaaS
- MCP's
- Application Security
- Incident Response
- DAST
- Security Event Management
- Malware Reverse Engineering
- Open Source Intelligence
- AI Security
- Threat Hunting
- Secure Code Review
- AWS Security
- Ethical Hacking
- Network Forensics
- Cloud Security

### Scripting

- Python
- PHP
- Bash
- JavaScript
- SQL

## Tools

- Coverity
- Burpsuite
- WinHex
- Ghidra
- IBM QRadar
- Linux
- Ettercap
- EnCase
- Black Bag
- Volatility
- Hashing
- Hydra
- Nessus
- Semgrep
- ZAP
- OllyDbg
- Snort
- Log Rhythm
- Nmap
- Aircrack-ng
- FTK
- X-Ways
- Tcpdump
- Write Blocker
- John the ripper
- Open Source Security Tools
- Docker
- OWASP
- IDA
- Zeek
- Metasploit
- Wireshark
- SQLMap
- Autopsy
- AXIOM
- Falcon
- Splunk
- Nikto